Multi-Scale Feature Extraction for Cognitive Load Classification

Quoc-Huy Nguyen[1][0000-0003-4996-628X], Quan Do Minh[1], Nhu-Tai Do[1][0000-0002-7034-0364], and Lien Nguyen Thi Kim[2,*]

[1] Saigon University, Vietnam
[2] Asia Eastern University of Science and Technology, Taiwan
nqhuy@sgu.edu.vn, doquan19062004@gmail.com, dntai@sgu.edu.vn, ntklien@mail.aeust.edu.tw
* Corresponding author: ntklien@mail.aeust.edu.tw

Abstract. In recent years, cognitive load detection has garnered increasing attention in the field of human–computer interaction, given its relevance to safeguarding mental health and enhancing work efficiency. This study investigates the detection of cognitive load states using physiological signals derived from the CogLoad dataset. We propose an approach that integrates the Simple Moving Average smoothing technique with Multiscale Analysis to extract discriminative features across varying temporal segments. Experimental evaluation demonstrates that the XGBoost model achieves an accuracy of 75.65% and an area under the ROC curve of 77%, confirming its effectiveness in classifying cognitive load states. These findings underscore the potential of real-time cognitive load monitoring to advance both user performance and overall well-being.

Keywords: Cognitive Load Detection · Physiological Signals · Human–Computer Interaction.

1 Introduction

Cognitive load refers to the amount of cognitive resources, such as attention, working memory, and decision-making capacity, etc., utilized during task performance [13]. It is a multidimensional construct that reflects the interaction between task characteristics and the cognitive demands imposed on the performer [9]. When cognitive load exceeds working memory capacity, performance deteriorates, and stress or anxiety may occur (overload). Conversely, excessively low cognitive load can lead to disengagement or boredom [13, 10]. In the field of education, measuring and managing cognitive load has long been recognized as essential for optimizing instructional design and improving learning outcomes [10]. More broadly, cognitive load assessment is increasingly relevant in domains such as healthcare, gaming, and human-computer interaction (HCI), where adaptive systems can respond to users' mental states in real time [1, 3].

The recent proliferation of wearable sensors and ubiquitous devices has created new opportunities for real-time monitoring of physiological signals related to cognitive load [11, 6]. Physiological indicators such as galvanic skin response (GSR), heart rate (HR), respiratory rate (RR), and skin temperature (Temp) have been shown to correlate strongly with mental workload [2, 3, 14]. For example, in the Cognitive Load Monitoring Challenge (CogLoad), data were collected from participants under resting and high-load task conditions, using wristband sensors with a sampling frequency of 1 Hz for 30-second windows [8, 6]. This benchmark dataset has since facilitated several machine learning studies aimed at improving cognitive load detection accuracy [11, 14, 7].

Recent research has highlighted several methodological directions. Feature engineering remains a key challenge, with studies exploring statistical measures (mean, standard deviation, kurtosis), temporal dynamics, and hybrid approaches [3, 2, 14]. In addition, advanced classification techniques, such as tree-based ensemble methods and explainable AI (e.g., SHAP analysis), have been proposed to enhance performance and interpretability [7]. These efforts collectively suggest that robust cognitive load detection requires a careful balance between signal preprocessing, feature extraction, and model complexity.

In this study, we build upon these insights by proposing a multiscale feature extraction approach to infer cognitive load states from the CogLoad dataset. Specifically, physiological signals are first smoothed using a simple moving average to mitigate short-term fluctuations. Statistical features are then extracted across multiple temporal scales, enabling the model to capture both fine-grained and coarse-grained signal variations. We compare the performance of single-scale and multiscale models, with evaluation based on classification accuracy and area under the ROC curve (AUC). Our findings demonstrate that the proposed method effectively improves cognitive load classification, while also reinforcing the potential of wearable-based monitoring systems in real-time applications for education, healthcare, and workplace productivity.

This paper makes the following key contributions: (1) proposing a multiscale feature extraction approach for cognitive load detection from physiological signals, leveraging statistical measures across multiple temporal scales; (2) applying a signal smoothing step using Simple Moving Average (SMA) to reduce short-term fluctuations and improve feature stability; (3) conducting a comparative evaluation of single-scale and multiscale models using the CogLoad dataset, demonstrating that the multiscale model achieves higher accuracy and AUC in binary classification tasks; and (4) discussing the potential applications of real-time cognitive load monitoring in education, healthcare, and workplace productivity.

2 Related Works

User monitoring provides significant opportunities for accurately measuring cognitive load (CL) in real time. Such information can be leveraged by interactive systems to deliver adaptive support and personalized feedback, thereby enhancing user performance. In recent years, a wide range of approaches have been

Table 1: Representative studies on cognitive load detection using biosignals.

Research	**Signals (Hz)**	**Window Length**	**Model**	**Accuracy**
Anusha et al. (2018)	ECG (64), GSR (1000), ST (1000)	30–300	LDA, KNN, QDA, SVM	0.97
Pettersson et al. (2020)	EEG (512), ECG (1000)	45	SVM, RF, XG-Boost	0.97
Fan et al. (2020)	ST (1), GSR (1), HR (1), RR (1)	360	SVM	0.80
Borisov et al. (2021)	ST (1), GSR (1), HR (1), RR (1)	30	Ensemble model	0.66
Tervonen et al. (2021)	ST (1), GSR (1), HR (1), RR (1)	30	XGBoost	0.67
Hong et al. (2023)	ST (1), GSR (1), HR (1), RR (1)	30	LightGBM	0.70

developed for CL detection, including implementations based on the benchmark CogLoad dataset.

Research on CL has spanned multiple domains. In multimedia and online learning, for example, scholars have investigated how learners process information, how multimedia materials affect cognitive demands, and how instructional designs can be optimized to prevent overload [9]. These studies emphasize that aligning content complexity with learners' cognitive capacity is critical to improving learning outcomes.

Table 1 highlights representative works on CL detection from biosignals. Anusha et al. [1] monitored work stress using EDA, ECG, and skin temperature (ST), achieving 97.1% accuracy on 60-second windows by combining EDA and ST features. Pettersson et al. [11] employed EEG and ECG signals in the Maastricht Acute Stress Test, reporting that SVM outperformed RF and XG-Boost in classifying relaxed versus stressed states. Fan et al. [3] investigated workload effects on EEG and ECG, obtaining 80% accuracy by applying PCA to ECG-derived features and training an SVM classifier.

While these studies report strong performance, most rely on high-frequency signals and lab-based equipment, which limits their applicability for everyday, wearable-based monitoring. To overcome this limitation, the CogLoad challenge was introduced (see Fig. 1), providing a public benchmark dataset collected with wrist-worn sensors.

Fig. 1: The Microsoft Band 2 wristband used for data collection in the CogLoad challenge.

Gjoreski et al. [6] reported results from 13 teams participating in the challenge, using 825 sessions of GSR, HR, RR, and ST signals from 23 participants. Features included statistical measures (mean, variance, skewness, kurtosis, median, sum) and frequency-domain indicators (e.g., HRV power spectral density ratios). Various feature selection methods, such as the maximum information coefficient, sequential backward search, and Gini-based ranking, were applied. Classifiers including SVM, RF, XGBoost, and LightGBM were tested, with ensemble methods achieving the strongest results. Borisov et al. [2] achieved the top performance by extracting simple but robust statistical descriptors (minimum, maximum, mean, standard deviation, skewness) from wristband data.

Recent work has extended this line of research with deep learning and multimodal approaches, such as dual-attention networks for multi-sensor fusion [5], hybrid CNN–RNN frameworks integrating physiological and behavioral signals [12], and robust cross-sensor detection methods [4]. These studies highlight the ongoing transition from handcrafted features toward hybrid and deep architectures, confirming the importance of robustness, scalability, and generalization in cognitive load detection.

3 Materials and Methods

3.1 Problem Overview

The proposed framework aims to classify cognitive load from physiological signals collected by wearable devices. Each recording session consists of four 30-second signals—respiratory rate (RR), heart rate (HR), galvanic skin response (GSR), and skin temperature (ST)—sampled at 1 Hz, resulting in 825 sessions from 23 participants [6]. Figure 2 illustrates the overall workflow of the approach. In the preprocessing stage, raw signals are smoothed using a median filter and a simple moving average (SMA) with sliding windows of size 1 or 3, which has been shown to reduce noise and highlight physiological trends in wearable-based studies [2, 7]. The preprocessed signals are then segmented into three temporal

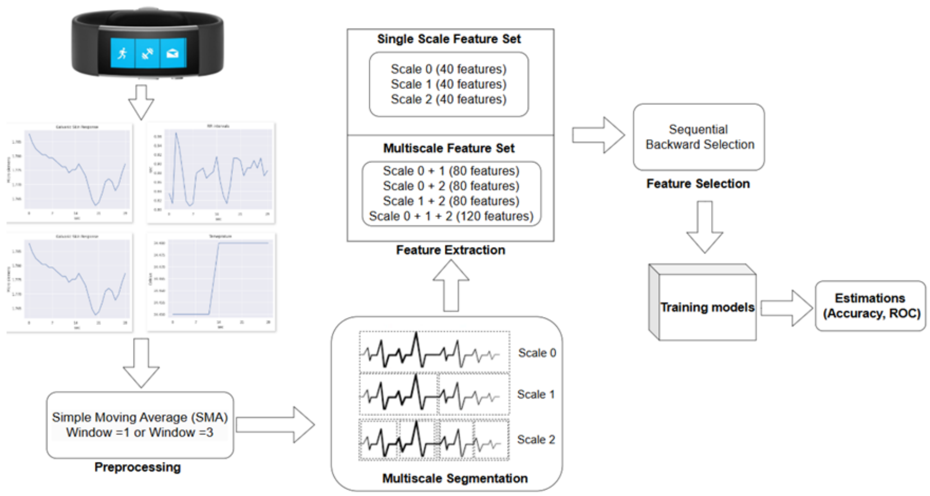

Fig. 2: Overall workflow of the proposed method for cognitive load detection based on multiscale feature extraction and machine learning.

levels (scales 0, 1, and 2), corresponding to different segment lengths, following multiscale strategies applied in previous cognitive load research [11, 3]. From each segment, 40 statistical features such as mean, variance, skewness, kurtosis, minimum, and maximum are extracted, producing single-scale feature sets with 40 features per level and multiscale feature sets of up to 120 features. To reduce redundancy and improve efficiency, a sequential backward feature selection algorithm is applied, which has been widely adopted for physiological feature reduction in cognitive state estimation [11]. All features are normalized using Min–Max scaling to ensure consistency across participants. The dataset is split into a training set comprising 632 sessions from 18 participants and a testing set containing 193 sessions from 5 participants [6].

For classification, six models are considered: Linear Discriminant Analysis (LDA), Random Forest (RF), Support Vector Machine (SVM), XGBoost, Multi-Layer Perceptron (MLP), and TabNet. These models represent a spectrum of traditional machine learning and modern deep learning approaches. Linear projection methods such as LDA have been used effectively in physiological signal classification [1], while ensemble tree-based methods like RF and boosting algorithms such as XGBoost are popular for handling structured tabular data [11]. SVM has been widely adopted in workload and stress detection due to its margin-based separation capabilities [3]. Neural networks such as MLP enable nonlinear representation learning, whereas TabNet represents a recent deep learning architecture optimized for tabular data with attention-based feature selection [7]. Hyperparameter optimization is conducted through GridSearchCV (applied to LDA, RF, SVM, and XGBoost), RandomizedSearchCV, RandomSearch, and Optuna, with GroupKFold cross-validation ($n = 3$) to account for inter-participant variability. Model performance is evaluated on the test set using two primary metrics: classification accuracy and the area under the ROC curve (AUC), which are commonly reported in cognitive load detection benchmarks [6, 2].

3.2 Experimental Data

The CogLoad dataset was collected from 23 participants using a commercially available wearable device worn on the non-dominant arm [6]. Each participant completed a series of cognitive load estimation tasks while their physiological signals were recorded, including respiratory rate (RR), heart rate (HR), galvanic skin response (GSR), and skin temperature (ST). A total of 825 recording sessions were obtained, where rest periods are labeled as "no load" (0) and task periods as "cognitive load" (1).

The cognitive load assessments consisted of two categories: the N-back test and six elementary cognitive tasks (ECTs). The N-back paradigm, administered at levels $n = 2$ and $n = 3$, is widely used to evaluate working memory capacity and induce varying degrees of cognitive demand. The six ECTs included Gestalt Completion, Finding Hidden Patterns, Finding A's, Number Comparison, Pursuit Test, and Scattered X's. Each task was designed with three levels of difficulty (easy, medium, and hard) to elicit different levels of cognitive load. After completing each task, participants rated their perceived workload using the NASA-TLX questionnaire, a standardized tool developed by NASA to measure six dimensions of workload: mental demand, physical demand, temporal demand, performance, effort, and frustration. To mitigate fatigue effects, participants were instructed to rest for three minutes following the completion of each questionnaire.

Each recording session consisted of four 30-second signals sampled at 1 Hz. Of the 825 sessions, 632 recordings from 18 participants were assigned to the training set, while 193 recordings from 5 participants were reserved for the test set. This balanced design, combining objective physiological measurements with subjective workload ratings, provides a robust foundation for evaluating cognitive load detection models.

3.3 Proposed method

Data Preprocessing Physiological signals collected from wearable devices are often affected by measurement errors and natural fluctuations, resulting in considerable noise. Such noise not only degrades data quality but can also obscure underlying trends, thereby hindering both signal analysis and the training of machine learning models.

To address this issue, a moving average filter is applied with window sizes of either one or three samples, corresponding to two levels of smoothing, as illustrated in Figure 3. This technique effectively suppresses high-frequency noise while preserving the essential characteristics of the physiological signals.

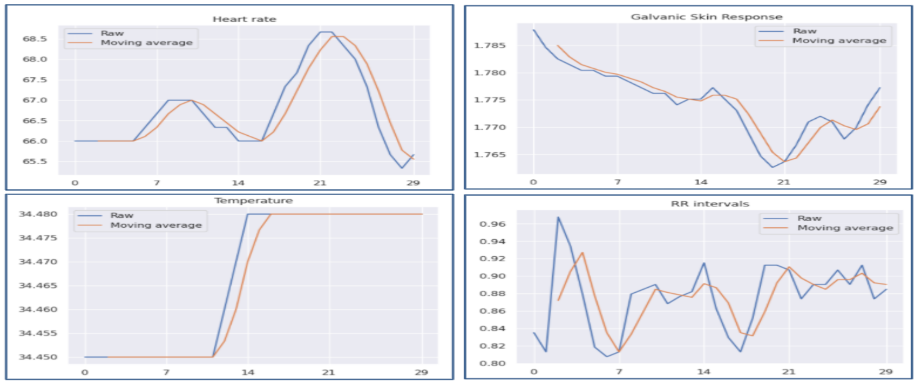

Fig. 3: Comparison of raw and smoothed signals after applying the moving average filter.

The preprocessing step enhances the stability and clarity of the data, facilitating more reliable extraction of both statistical and temporal features. Consequently, the improved signal quality contributes to better discrimination of physiological states and enhances the performance of machine learning models in cognitive load classification tasks.

Signal Segmentation by Time Scale To capture both global and local characteristics of the physiological signals, a multiscale segmentation strategy is employed. The core idea is to divide each preprocessed signal into progressively finer segments, enabling the extraction of statistical features at different temporal levels. This allows the model to learn both short-term variations and long-term dependencies, which are crucial for distinguishing cognitive load states. The segmentation scheme is illustrated in Figure 4.

At the first level (scale 0), the entire signal

$$x = [x_1, x_2, \ldots, x_n]$$

is treated as a single block, and a feature function $f(\cdot)$ is applied to obtain:

$$y_1^{(0)} = f(x_1, x_2, \ldots, x_n).$$

At the second level (scale 1), the signal is divided into two equal halves, each processed independently to generate local features:

$$y_1^{(1)} = f(x_1, \ldots, x_{n/2}), \quad y_2^{(1)} = f(x_{n/2+1}, \ldots, x_n).$$

At the third level (scale 2), each segment from scale 1 is further subdivided based on a scaling factor $\alpha \in (0, 1)$. For a segment of length $n/2$, the subdivision lengths are defined as:

$$m_1 = \left\lfloor \frac{n}{2} \cdot \alpha \right\rfloor, \quad m_2 = \left\lfloor \frac{n}{2} \cdot (1 - \alpha) \right\rfloor.$$

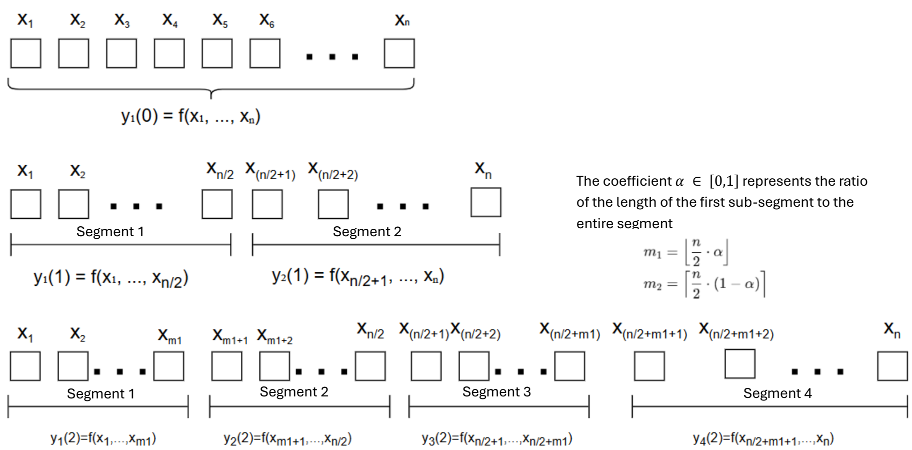

Fig. 4: Signal segmentation across three temporal scales (scale 0, scale 1, and scale 2). The multiscale approach enables feature extraction from both global and local portions of the signal.

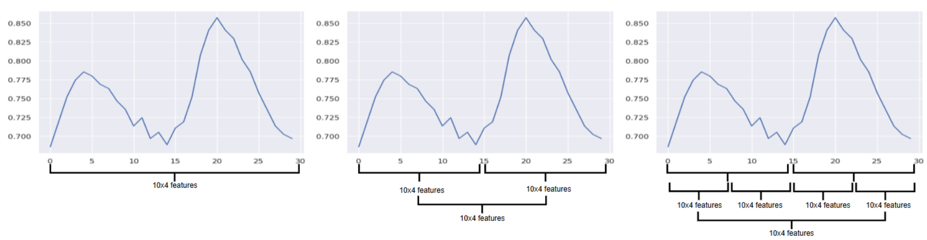

Fig. 5: Single-scale feature extraction at three temporal levels: scale 0, scale 1, and scale 2.

This generates four smaller sub-segments, producing feature outputs

$$y_1^{(2)}, y_2^{(2)}, y_3^{(2)}, y_4^{(2)},$$

which represent the beginning and end portions of the two halves.

By integrating multiple time scales, the model can capture complementary patterns in the data: scale 0 reflects global trends, scale 1 captures medium-range structures, and scale 2 emphasizes fine-grained variations. This hierarchical segmentation offers a more comprehensive feature representation for machine learning models, thereby enhancing the accuracy of cognitive load classification.

Feature Extraction Feature statistics provide essential information about the characteristics of physiological signals. The mean reflects the central tendency, while the standard deviation measures dispersion around the mean. Skewness indicates asymmetry of the distribution, and kurtosis quantifies its sharpness. The

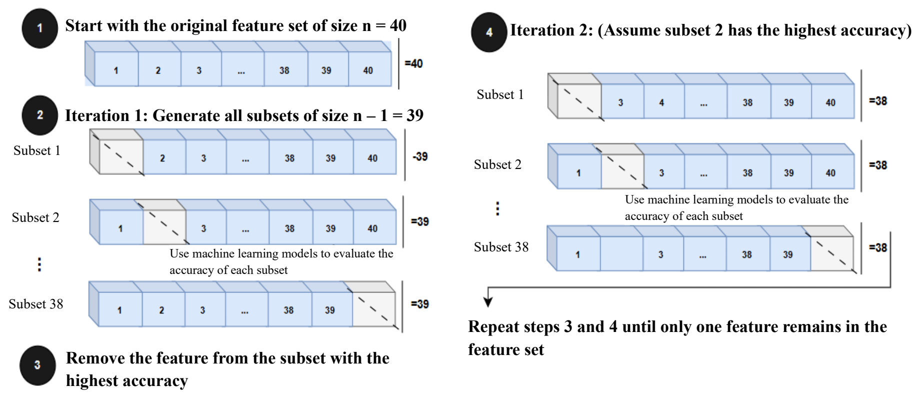

Fig. 6: Performance of single- and multi-scale feature sets in cognitive load classification; combining scales improves accuracy.

average of the first- and second-order derivatives captures short-term variability and signal dynamics, whereas the disparity measure (diff2) characterizes uneven fluctuations. Percentile-based features, such as the 25th and 75th percentiles, describe the spread of the distribution, and the interquartile range (IQR) highlights the concentration of values around the median. The range, defined as the difference between maximum and minimum values, provides insight into overall signal amplitude. Together, these descriptors yield a comprehensive view of global and local signal properties and be highly effective in stress and workload detection tasks [1, 3, 2].

To capture both short- and long-term fluctuations, a multiscale analysis is applied to the RR, HR, GSR, and ST signals at three temporal levels (Figure 5). At scale 0, features are extracted directly from the original 30-second signals (10×4 features). At scale 1, each signal is divided into two equal 15-second segments, and features are computed for each segment before averaging (10×4 features). At scale 2, the signal is further divided into four equal sub-segments of approximately 7.5 seconds, from which features are extracted and averaged (10×4 features). Specifically, scale 0+1, scale 0+2, scale 1+2 (each with 80 features), and scale 0+1+2 (120 features) are the combinations that are created. Recent research has revealed that integrating features from various time periods greatly enhanced workload and stress detection classification performance [7].

4 Experiments and Discussion

4.1 Model Evaluation

We can evaluate the effects of noise reduction and temporal segmentation on model performance using these configurations.

Accuracy reflects the proportion of correctly classified samples among the total number of samples and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

where TP (true positives) are correctly classified positive samples, TN (true negatives) are correctly classified negative samples, FP (false positives) are negative samples misclassified as positive, and FN (false negatives) are positive samples misclassified as negative. Since the CogLoad dataset is approximately balanced between the "rest" and "task" states, accuracy serves as a suitable and unbiased performance measure. Unlike in imbalanced datasets, where accuracy can be misleading, it reliably reflects the overall predictive capability across both classes here.

To complement accuracy, the AUC metric captures the discriminative ability of models independent of any specific threshold. The AUC is derived from the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR), defined respectively as:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

The AUC value ranges from 0.5 (no discriminative ability, equivalent to random guessing) to 1.0 (perfect discrimination). A higher AUC indicates greater robustness in distinguishing between cognitive load states, which is especially important in real-world applications where classification thresholds may vary depending on the context.

By combining accuracy and AUC, the evaluation framework ensures comprehensiveness: accuracy provides an intuitive measure of correct classifications under balanced data conditions, while AUC reflects threshold-independent discriminative power. Recent cognitive load detection studies using wearable sensors confirm the importance of this two-metric strategy [2, 4], validating its appropriateness for benchmarking machine learning models in cognitive state classification tasks.

4.2 Experiments and Discussion

Experiments with Single-Scale Features Table 2 reports the results of single-scale experiments without applying signal smoothing. Overall, the machine learning models still exhibit reasonable classification performance. Among them, XGBoost (XGB) achieves the highest accuracy of 0.7504 at scale 1, demonstrating its ability to capture discriminative patterns from raw signals effectively. Linear Discriminant Analysis (LDA) and Random Forest (RF) also yield stable results, with ROC values consistently in the range of 0.74–0.75. In contrast, the Support Vector Machine (SVM) performs poorly at scale 1, with an ROC of only 0.25, indicating instability of the extracted features under unsmoothed conditions.

Table 2: Model performance with single-scale features (without SMA).

	Scale 0		Scale 1		Scale 2	
Model	Accuracy	ROC	Accuracy	ROC	Accuracy	ROC
LDA	0.7150	0.74	0.7150	0.73	0.6943	0.74
SVM	0.7047	0.75	0.7098	0.25	0.6580	0.71
RF	0.7409	0.75	0.7254	0.75	0.6995	0.70
XGB	0.7254	0.74	0.7504	0.76	0.7047	0.73
MLP	0.6943	0.73	0.7047	0.76	0.6788	0.76
TabNet	0.6166	0.57	0.6114	0.63	0.6010	0.56

When signal smoothing with SMA (window = 3) is applied, as shown in Table 3, the classification performance improves noticeably across most models. For example, LDA at scale 2 increases its ROC from 0.74 to 0.75, and SVM at scale 1 shows a dramatic gain from 0.25 to 0.73, underscoring the stabilizing effect of smoothing on sensitive models. In terms of accuracy, Random Forest achieves the best result with 0.7409 at scale 0, confirming its robustness and ability to effectively leverage smoothed features.

Table 3: Model performance with single-scale features (with SMA, window = 3).

	Scale 0		Scale 1		Scale 2	
Model	Accuracy	ROC	Accuracy	ROC	Accuracy	ROC
LDA	0.7150	0.76	0.7047	0.72	0.7150	0.75
SVM	0.7047	0.76	0.6891	0.73	0.7254	0.76
RF	0.7409	0.74	0.7306	0.75	0.7095	0.75
XGB	0.7202	0.75	0.7098	0.73	0.7150	0.70
MLP	0.7150	0.76	0.6839	0.70	0.6839	0.72
TabNet	0.6512	0.66	0.6166	0.66	0.5959	0.63

These findings highlight that both scale selection and the application of smoothing techniques substantially influence model performance. Nevertheless, relying solely on a single scale may limit the exploitation of complementary temporal information. A more promising direction is to integrate features across multiple scales to maximize discriminative power, which will be explored in the next section on multiscale experiments.

Experiments with Multi-Scale Features Table 4 presents the results of multi-scale experiments without applying signal smoothing. Among all models, Random Forest (RF) achieved the best performance with an accuracy of 0.7461 and an ROC of 0.76 at the scale combination 0+1. This indicates that RF is capable of leveraging complementary information across scales, even in the absence of preprocessing. XGBoost (XGB) and Linear Discriminant Analysis (LDA) also performed competitively, each achieving an accuracy of 0.7358 at the 0+2 and 0+1+2 combinations, respectively.

By contrast, the Multilayer Perceptron (MLP) and TabNet models showed limited effectiveness under non-smoothed conditions. Notably, TabNet produced the weakest results, with an accuracy of only 0.6010 and ROC of 0.58 at the 1+2 scale, suggesting that attention-based deep models are less robust when applied directly to raw multi-scale features.

Table 4: Model performance with multi-scale features (without SMA).

	Scale 0+1		Scale 0+2		Scale 1+2		Scale 0+1+2	
Model	Acc	ROC	Acc	ROC	Acc	ROC	Acc	ROC
LDA	0.7254	0.75	0.7358	0.77	0.6995	0.74	0.7254	0.76
SVM	0.7254	0.76	0.7090	0.73	0.7202	0.77	0.7098	0.73
RF	0.7461	0.76	0.7358	0.73	0.7358	0.74	0.7358	0.76
XGB	0.7358	0.75	0.7254	0.75	0.7202	0.69	0.7254	0.73
MLP	0.7150	0.77	0.6891	0.73	0.7098	0.74	0.7202	0.76
TabNet	0.6218	0.61	0.6529	0.67	0.6010	0.58	0.6425	0.64

When signal smoothing with SMA (window = 3) is introduced, performance improves across nearly all models, as summarized in Table 5. The XGB model achieved the overall best result, with an accuracy of 0.7565 and ROC of 0.77 at scale 0+2, confirming its strong capacity for feature learning and information synthesis under smoothed multi-scale inputs. RF maintained robust and consistent results, with accuracy = 0.7513 and ROC = 0.76 at scale 0+1. Interestingly, the MLP model achieved the highest ROC value (0.78) at scale 0+2, highlighting the benefit of smoothing for stabilizing deep neural models in this setting.

TabNet, while still underperforming compared with traditional methods, showed moderate improvement with smoothing, suggesting partial mitigation of its sensitivity to noisy features.

Table 5: Model performance with multi-scale features (with SMA, window = 3).

	Scale 0+1		Scale 0+2		Scale 1+2		Scale 0+1+2	
Model	Acc	ROC	Acc	ROC	Acc	ROC	Acc	ROC
LDA	0.7150	0.73	0.7254	0.75	0.7047	0.74	0.7254	0.74
SVM	0.7098	0.76	0.7202	0.25	0.7098	0.74	0.7306	0.76
RF	0.7513	0.76	0.7407	0.76	0.7409	0.74	0.7306	0.71
XGB	0.7150	0.74	0.7565	0.77	0.7358	0.75	0.7150	0.72
MLP	0.7358	0.77	0.7409	0.78	0.6943	0.75	0.7150	0.76
TabNet	0.6218	0.61	0.6321	0.59	0.6425	0.68	0.6425	0.68

These results demonstrate that combining features across multiple temporal scales enhances the discriminative power of the models compared to single-scale analysis. Moreover, the application of SMA further stabilizes feature representations, particularly benefiting sensitive models such as SVM and deep networks like MLP. Overall, ensemble tree-based methods (RF, XGB) consistently deliver the most reliable performance across different experimental settings.

4.3 Comparison with Related Studies

To situate the effectiveness of the proposed method in a broader context, we compared it against machine learning approaches reported in the *UBICOMP 2020 Cognitive Load Classification Challenge*, a widely recognized benchmark in this field. The competition attracted diverse solutions ranging from traditional classifiers such as Logistic Regression (LR), Support Vector Machines (SVM), and Random Forests (RF) to more complex ensemble and neural network–based models.

As summarized in Table 6, the proposed method—combining multi-scale feature extraction, signal smoothing using SMA, and classification with XGBoost—achieved the highest accuracy of 0.7565. This result surpasses all published methods from the UBICOMP 2020 competition, including the top-performing ensemble of gradient boosting decision trees (accuracy = 0.694). Notably, several advanced models such as convolutional and recurrent neural networks yielded lower performance (0.560 and 0.554, respectively), highlighting the challenges of directly applying deep architectures to small-scale, noisy physiological datasets without tailored preprocessing or feature engineering.

Table 6: Comparison between UBICOMP 2020 competition methods and the proposed approach.

Method	**Accuracy**
Ensemble of 7 Gradient Boosting Decision Trees	0.694
Support Vector Machine	0.679
Ensemble of Support Vector Machines	0.674
Logistic Regression	0.663
Random Forest	0.653
Weighted Sum of Individualized and Global LR	0.653
Multilayer Perceptron	0.648
Logistic Regression (variant)	0.648
Support Vector Machine (variant)	0.627
XGBoost Classifier	0.580
Convolutional Neural Network	0.560
Recurrent Neural Network	0.554
Logistic Regression (baseline)	0.503
Proposed Method (Multi-Scale + SMA + XGBoost)	**0.7565**

These findings confirm that the integration of multi-scale feature engineering and SMA-based preprocessing with XGBoost offers a robust and effective solution for cognitive load detection from wearable sensor data. Beyond outperforming baseline methods, this approach demonstrates the value of combining statistical signal analysis with modern ensemble learning, providing both higher accuracy and stronger generalizability.

5 Conclusion

This study introduced a novel framework for cognitive load classification from physiological signals by integrating moving average–based signal smoothing, temporal segmentation, and multi-scale statistical feature extraction. Experimental results demonstrated that the proposed approach, when combined with the XGBoost classifier, achieved the best performance with an accuracy of 75.65% and an AUC of 0.77. These results not only surpass all reported methods from the CogLoad competition but also highlight the importance of coupling signal preprocessing with multi-scale analysis to enhance discriminative power in cognitive state recognition.

The findings confirm that statistical multi-scale features, supported by smoothing techniques, can provide robust representations for physiological time-series data. Beyond achieving competitive performance, the proposed method opens up promising avenues for real-time cognitive load monitoring using wearable devices. Such applications hold significant potential for improving user well-being and optimizing performance in education, healthcare, and human–computer interaction.

References

1. Anusha, A., Jose, J., Preejith, S., Jayaraj, J., Mohanasankar, S.: Physiological signal based work stress detection using unobtrusive sensors. Biomedical Physics & Engineering Express **4**(6), 065001 (2018)
2. Borisov, V., Kasneci, E., Kasneci, G.: Robust cognitive load detection from wristband sensors. Computers in Human Behavior Reports **4**, 100116 (2021)
3. Fan, X., Zhao, C., Zhang, X., Luo, H., Zhang, W.: Assessment of mental workload based on multi-physiological signals. Technology and Health Care **28**(1_suppl), 67–80 (2020)
4. Foltyn, A., Deuschel, J., Lang-Richter, N.R., Holzer, N., Oppelt, M.P.: Evaluating the robustness of multimodal task load estimation models. Frontiers in Computer Science **6**, 1371181 (2024)
5. Gao, W., Zhang, L., Teng, Q., He, J., Wu, H.: Danhar: Dual attention network for multimodal human activity recognition using wearable sensors (2021), `https://arxiv.org/abs/2006.14435`
6. Gjoreski, M., Mahesh, B., Kolenik, T., Uwe-Garbas, J., Seuss, D., Gjoreski, H., Luštrek, M., Gams, M., Pejović, V.: Cognitive load monitoring with wearables–lessons learned from a machine learning challenge. IEEE access **9**, 103325–103336 (2021)

7. Kim, J.K., Lee, K., Hong, S.G.: Cognitive load recognition based on t-test and shap from wristband sensors. Hum. Centric Comput. Inf. Sci **27**, 13 (2023)
8. Li, X., De Cock, M.: Cognitive load detection from wrist-band sensors. In: Adjunct Proceedings of the 2020 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2020 ACM International Symposium on Wearable Computers. pp. 456–461 (2020)
9. Mayer, R.E.: The Cambridge handbook of multimedia learning. Cambridge university press (2005)
10. Paas, F., Renkl, A., Sweller, J.: Cognitive load theory and instructional design: Recent developments. Educational psychologist **38**(1), 1–4 (2003)
11. Pettersson, K., Tervonen, J., Närväinen, J., Henttonen, P., Määttänen, I., Mäntyjärvi, J.: Selecting feature sets and comparing classification methods for cognitive state estimation. In: 2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE). pp. 683–690. IEEE (2020)
12. Siddiqui, S.T.: Hybrid cnn-rnn deep learning framework for eeg-based mental health disorder diagnosis with explainable ai. In: 2025 International Conference on Intelligent and Cloud Computing (ICoICC). pp. 1–6. IEEE (2025)
13. Sweller, J.: Cognitive load theory. In: Psychology of learning and motivation, vol. 55, pp. 37–76. Elsevier (2011)
14. Tervonen, J., Pettersson, K., Mäntyjärvi, J.: Ultra-short window length and feature importance analysis for cognitive load detection from wearable sensors. Electronics **10**(5), 613 (2021)